

8.3 LoRA, Adapters, and Federated Updates

A low-rank pair BA instead of rewriting W . Small deltas you can ship or keep local.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Full fine-tuning moves every entry of W . **LoRA** (Hu et al., 2021) freezes W and learns a **low-rank residual**. You store a few megabytes per task instead of a second copy of the 7B.

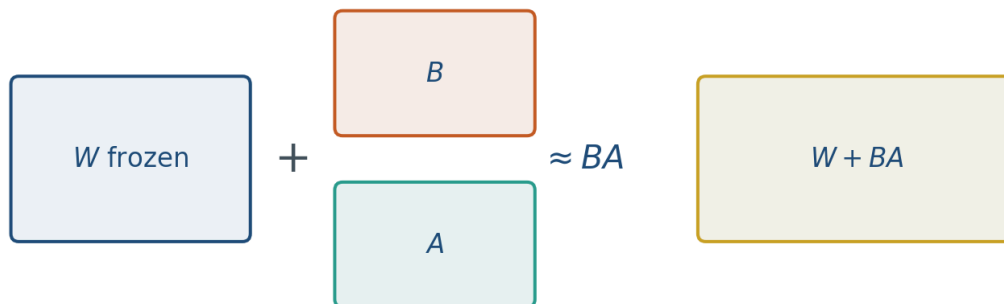
1. $W + BA$

For a linear map $W \in \mathbb{R}^{d \times k}$,

$$h = Wx + \frac{\alpha}{r}BAx,$$

with $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, $\text{rank } r \ll \min(d, k)$. Initialize $B = 0$ so the adapter starts as the base model. Train only A, B (and maybe a bias). Attention projections and MLP up/down maps are the usual insertion points.

LoRA: train a thin pair BA , leave W alone



QLoRA quantizes W to 4-bit and still trains float adapters. Same algebra. Lab 8 uses a 4×4 frozen W so you can print BA .

2. Rank, merge, many tasks

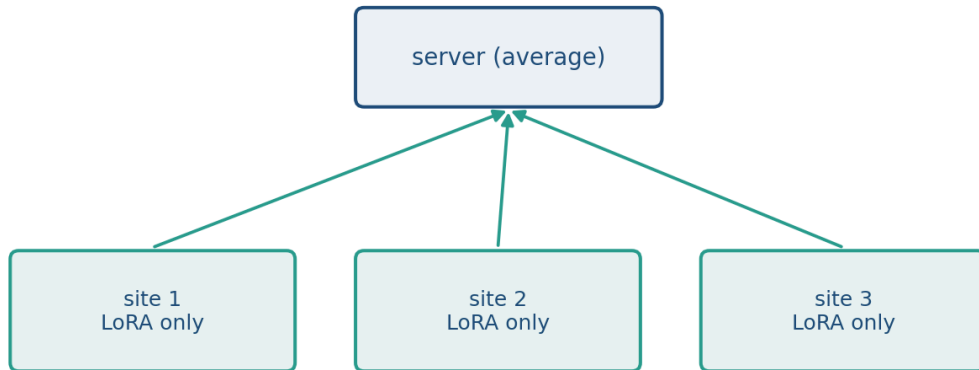
Small r (4, 8, 16) is often enough for style and instruction shifts; harder domain shifts may want larger r or more layers. At deploy time you can **merge** $W \leftarrow W + (\alpha / r)BA$ and throw the adapter away, or keep several adapters and swap them.

One adapter per skill is how you avoid the XOR-then-AND overwrite (note **8.2**): the base stays, the skill is a file.

3. Federated adapters

Sites that cannot ship raw text can still train a LoRA locally and send A, B to a server. The server averages adapters (or stacks them) without seeing documents. That is **federated** fine-tuning in the sense this course needs: communication is the adapter, not the corpus. Privacy is not automatic (adapters can leak), but the footprint is the right size.

Federated sketch: ship adapters, not raw documents



For a project: report r , which modules you adapted, α , and whether you merged. Compare to a small full fine-tune if you can afford it.

4. Practice

1. If $r = 1$ and $d = k = 4$, how many trainable numbers are in (A, B) versus W ?
2. Why initialize $B = 0$ rather than $A = 0$?
3. A hospital trains a LoRA and sends it to you. What did they *not* send, and what can still leak?